ASUS UGEN AI LEAGUE • BATTLEFIELD LIGHTNING | GLOBAL ONLINE TRACK

AegisEdge

Privacy-First Workplace Safety Guardian on ASUS UGen300 USB

THEME: Workplace AI | TRACK: Battlefield Lightning (Global Online Track)

TARGET HARDWARE: ASUS UGen300 USB AI Accelerator (UGen300-USB-8G)

TEAM: refrigerator1

Workplace Safety & Inspection Challenges

Objective: Automated, zero-delay compliance verification across high-risk sites.

Industrial Realities

- Manufacturing floors and construction sites present severe hazard risks.
- Continuous PPE compliance (helmets, high-vis vests) and restricted zone monitoring are critical.
- Human manual oversight is prone to fatigue, distraction, and coverage gaps.

Monitoring Imperative

- Traditional passive recording cameras only document accidents AFTER they happen.
- True hazard prevention demands real-time, sub-second proactive warnings.
- Automated vision systems must operate non-stop with near-zero latency.

02 / CRITICAL BOTTLENECKS

Why Cloud AI Fails in Workplace Safety

Conclusion: Safety vision processing must transition strictly to localized Edge NPUs.

01. PRIVACY RISK

Worker Resistance

Continuous cloud video streaming violates worker union privacy standards and risks exposing sensitive site IP or operational data.

02. LATENCY LAG

Network Delays

Bandwidth congestion and network jitter cause multi-second delays in alert signals, rendering hazard warnings useless.

03. OPERATIONAL COST

Heavy Subscriptions

Recurring cloud GPU server costs and high bandwidth requirements make multi-camera setups financially unfeasible for SMEs.

03 / CORE QUESTION

The Fundamental Problem Statement

Resolving the tension between privacy, speed, and real-time accuracy.

“ How do we achieve real-time, high-accuracy PPE compliance without sending a single frame of video off-site? ”

ANSWER:

100% On-Device Neural Processing via ASUS UGen300 USB AI Accelerator (UGen300-USB-8G).

AegisEdge Solution Architecture

Core Principle: Privacy by design, safety through localized speed.

KEY HIGHLIGHT: Employs USB 3.2 connectivity to instantly bring 40 TOPS edge AI capabilities to existing factory host PCs and IPCs without host hardware teardown.

INFERENCE

On-Device Acceleration

40 TOPS Hailo-10H execution handles high-FPS PPE detection natively on site.

PRIVACY

Zero Data Transmission

Raw video is destroyed immediately in volatile memory. Only non-person metadata exits.

RESPONSE

Instant Interlock

Hardware relays trigger local physical alarms and safety interlocks under 15ms.

Model Choice & Optimization Strategy

Leveraging Hailo Model Zoo pre-trained weights for rapid deployment.

Lightweight YOLO Architecture

• Fine-tuned YOLOv8-Nano / YOLOv9-Tiny optimized specifically for safety equipment detection.

• High throughput frame processing engineered for multi-stream input.

• Target Classes: [Helmet, Vest, Person, Danger Zone].

Quantization & Compilation

• Utilizing Hailo Dataflow Compiler to quantize PyTorch/ ONNX models into Hailo Executable Format (.hef).

• Format: INT4 / INT8 Optimization for 8GB LPDDR4x Memory.

• Minimizes memory footprint while preserving detection mAP.

06 / SYSTEM STACK

Hardware & Software Integration

Zero external software dependencies for off-grid factory deployment.

HARDWARE ACCELERATOR

ASUS UGen300-USB-8G

Hailo-10H NPU (40 TOPS INT4) with 8GB dedicated LPDDR4x over USB 3.2 Gen2 interface.

RUNTIME / SDK

HailoRT API

C++/Python asynchronous inference engine providing efficient model memory management.

STREAM PIPELINE

OpenCV RTSP

High-efficiency multi-camera RTSP video ingestion and frame pre-processing pipeline.

LOCAL DASHBOARD

FastAPI UI

Lightweight Web UI with WebSockets for real-time local alert logging and monitoring.

End-to-End Processing Pipeline

Entire pipeline operates locally within a closed factory intranet.

HARDWARE FLOW: IP Camera (RTSP) → Edge Host PC → [USB 3.2] → ASUS UGen300-USB-8G (Hailo-10H NPU) → Real-time Alert Engine

STAGE 1

RTSP Camera

Multi-stream IP camera video ingestion over factory LAN.

STAGE 2

ASUS UGen300-USB-8G

Hailo-10H NPU accelerated neural inference over USB 3.2.

STAGE 3

Safety Engine

Spatial geofence calculation and PPE compliance logic.

STAGE 4

Local Action

Instant local alarm triggers, WebSockets, and event log audit.

ASUS UGen300 USB Advantage

Dedicated 8GB onboard memory accommodates complex vision pipelines effortlessly.

Zero-Teardown Upgrade

The UGen300-USB-8G enables instant AI augmentation over USB 3.2, removing the need for internal PCIe/M.2 slot installation or custom carrier boards.

Dedicated 8GB LPDDR4x Memory

The on-module 8GB LPDDR4x memory easily handles multi-class PPE detection models and high-resolution stream frames directly on the accelerator.

40 TOPS

INT4 CAPACITY

Hailo-10H High Throughput

100%

LOCAL PRIVACY

Zero Video Cloud Egress

<15ms

END-TO-END LATENCY

Instant Alarm Trigger

4-Step Deployment Strategy

Ensuring zero accuracy loss post INT8 quantization.

STEP 01

Model Training

Train YOLO on custom workplace safety gear datasets in PyTorch.

STEP 02

ONNX Export

Export trained weights to ONNX format with fixed input dimensions.

STEP 03

Hailo Quantize

Compile ONNX model into .hef binary using Hailo Dataflow Compiler.

STEP 04

Stage II Execution

Deploy natively on physical ASUS UGen300-USB-8G hardware.

Impact on Manufacturing & Construction

Democratizing enterprise-grade AI safety for small-to-medium factories.

01. COST-EFFECTIVE DEPLOYMENT

Avoids costly industrial host replacements by utilizing the USB-based UGen300-USB-8G on current factory infrastructure.

02. LOW LATENCY & HIGH PRIVACY

Edge inference over USB delivers sub-10ms logic response times while keeping raw video data strictly local.

03. COMPLETE WORKER TRUST

Eliminates facial recognition or cloud video storage, fully satisfying union privacy demands.

Stage II Tangible Deliverables

All source code, setup scripts, and benchmarks will be committed to GitHub.

DELIVERABLE 1

Compiled .hef Binary Model

Custom quantized safety gear detection model targeting Hailo-10H NPU architecture.

DELIVERABLE 2

Python HailoRT Application

Asynchronous pipeline managing multi-stream ingestion and detection logic.

DELIVERABLE 3

Local Monitoring Dashboard

Lightweight Web UI displaying live bounding boxes and event audit logs.

Technical Documentation & Sources

Repository updated regularly with dataset preparation scripts.

Hardware & SDK Specs

- ASUS UGen300: USB AI Accelerator Specs & Hailo-10H Data Sheet
- HailoRT Driver: Hailo Runtime C++/Python API Documentation
- Hailo Model Zoo: Pre-trained Object Detection Models

Datasets & Benchmarks

- Roboflow PPE: Workplace Safety Gear Detection Dataset
- Open Images V7: Public Construction Hazard Annotations
- YOLO Benchmarks: Ultralytics ONNX Export Guidelines

PROJECT REPOSITORY & SUBMISSION

AegisEdge

Empowering Everyday Safety with Edge AI on ASUS UGen300 USB

GitHub: https://github.com/refrigerator1/AegisEdge-UGen300

Thank you for your consideration • ASUS UGen AI League 2026 • Team: refrigerator1